AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

Biology Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL EIGHT questions. Questions 6, 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 7 Marks)

a. How does the following help a plant to control the rate of transpiration? (2 Marks)

i. Intercellular spaces

_____

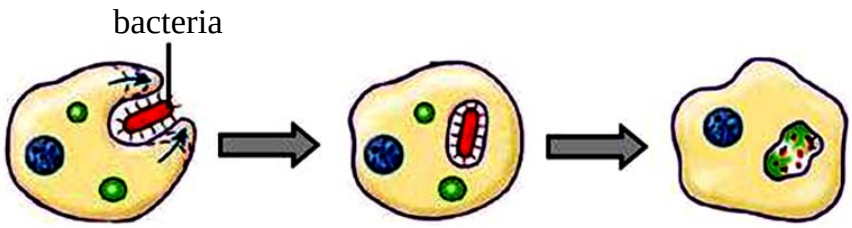
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ii. More stomata on the lower surface of a leaf

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b. The diagram shows the function of a type of blood cell.



i. Name the process. (1 Mark)

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ii. Give an account of the process shown in the diagram. (2 Marks)

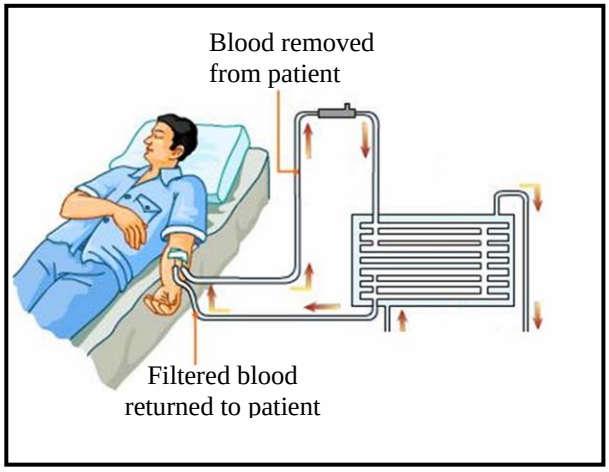
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c. The diagram shows a patient taking a dialysis treatment. (2 Marks)

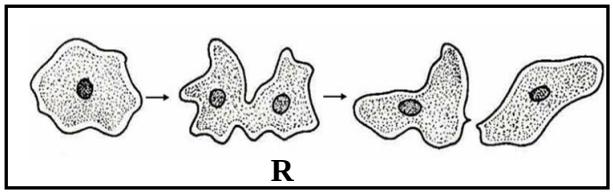


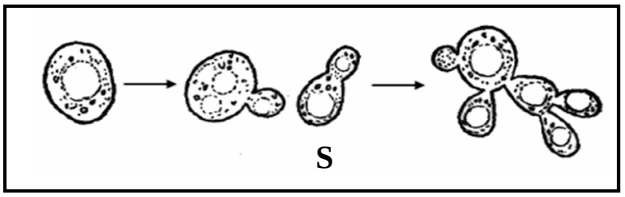
How are the concentrations of salts and urea set in the dialysis fluid to make it function like kidneys?

Q.2. (Total 5 Marks)

a. What happens to the ovary after fertilization? Give any TWO key events. (2 Marks)

b. Identify the types of asexual reproduction. Describe any ONE of them. (3 Marks)





R: _____ **S:** _____

Q.3. (Total 4 Marks)

a. Explain the genetic difference which determines sex in humans. (2 Marks)

b. How is the formation of transgenic animals beneficial for mankind? Give any TWO examples. (2 Marks)

Q.4. (Total 4 Marks)

According to a theory of evolution, "The variants better adapted to the environment are successful."

i. Name the scientist who proposed this theory. (1 Mark)

ii. Give THREE features of the animal (shown below) which make it adaptable to the polar environment. (3 Marks)



Q.5. (Total 5 Marks)

a. Draw a pyramid of number for the following food chain. (2 Marks)

Tree → grass hoppers → frogs → eagles

Space for diagram

b. Being a responsible member of a society, what direct role will you play to preserve the natural resources? List any THREE. (3 Marks)

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Q.6. (Total 5 Marks)

EITHER

a. Using the following key words, explain in detail the functions of blood.

Respiratory gases	hormones	temperature	excretory products	nutrients

OR

b. How does the structure of the following parts of a nephron help them to carry out their function effectively? Describe in detail.

i. Glomerulus ii. Renal tubules

Q.7. (Total 5 Marks)

EITHER

a. 'Locomotion helps animals in the continuation of their species.' Give FOUR reasons to justify the given statement.

OR

b. Explain, with examples, the contribution of environment, inadequate diet and inheritance in promoting eye diseases or disorders.

Q.8. (Total 5 Marks)

EITHER

a. Describe the interaction of biotic and abiotic components of an ecosystem. Give THREE examples to show the dependence of biotic components on abiotic.

OR

b. How is organic sewage treated in order to be changed into harmless liquid? Explain.

END OF PAPER